



Snake Lab

Coding Camp for Kids · 3 days + bonus Saturday

Build, code, and battle your own snake game.

In one week, kids go from playing a classic single-player snake to writing a JavaScript bot that competes head-to-head against their friends' bots in a tournament. They **build with AI** (Claude as a coding partner), **think like AI** (design strategies as tunable numbers), and **watch AI learn** (a tiny bot that gets smarter every round). Real code. Real servers. Real games they can keep playing at home.

What kids learn, day by day

Day 1 — Foundations

Morning: build a classic snake. Discover that 'the snake' is just a list in memory; what a game loop is. Afternoon: multiplayer — meet servers, WebSockets, and the bot's brain. Watch kids' join messages roll in live on the instructor's terminal.

Day 2 — Bots + AI

Morning — Hello, AI: meet your coding partner (Claude / ChatGPT). Kids ask AI to explain code, find bugs, and write a first draft together. Afternoon — Strategy: tune a real bot by hand using four weights (food, safety, blocking, open space).

Day 3 — Learning + Compete

Morning: watch a self-learning bot improve over rounds (Q-learning — the real idea behind reinforcement learning). Afternoon: tournament. Best bot wins. Parents invited.

Saturday (bonus) — Family Day

King-mode (snake eats snake) and fog-of-war tournaments. Parents play too. Kids show off the code they wrote on Day 3.

What every kid leaves with

- Three games they helped build, live online so they can keep playing at home
- A working JavaScript bot they wrote themselves — and one a Q-learning bot 'trained'
- Hands-on experience using AI (Claude / ChatGPT) as a coding partner — not as a magic-answer button
- Intuition for how AI learns: state, action, reward, repeat. The real loop, demystified.
- Confidence that 'code' is something they can write — not just consume

AGES

9 and up

DATES

To be announced (first trial run)

BRING

Just an iPad or laptop

CLASS SIZE

Up to 8 kids

PRICE

\$120 – \$150 · lunch included

LOCATION

TBA

PLEASE NOTE

- This is a **small, private kids activity camp** organized for friends — not a public program.
- Parents sign a simple waiver and provide emergency contact + allergy information.
- Spots are limited to keep things safe and manageable.

Interested?

Email yancyqin@gmail.com to hold a spot or ask questions.